

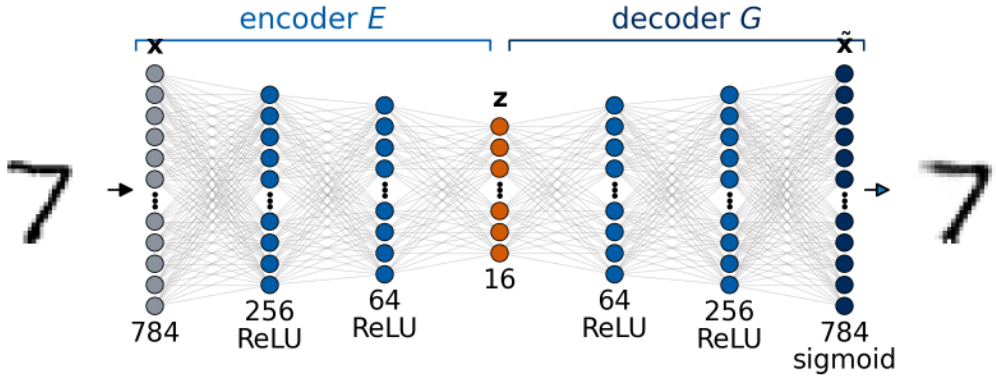
Lecture 6: Generative Models
Variational and Conditional Autoencoders
Generated Spins and the Hamiltonian

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

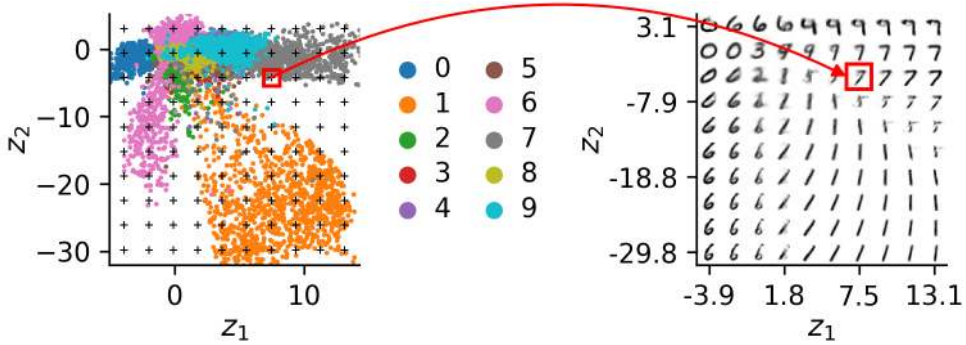
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The Autoencoder Compresses Each Image to a Point



Autoencoder (AE), $\mathbf{z} = E(\mathbf{x}) \in \mathbb{R}^d$ with $d \ll 784$, and $\tilde{\mathbf{x}} = G(\mathbf{z})$, trained with the mean squared error $\frac{1}{n}\sum_i \frac{1}{784}\|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$. Each image becomes one point of the latent space.

Decoding a Grid of Latent Points



- AE, $d = 2$; left, test codes $E(\mathbf{x})$ and a grid; right, the cell at (z_1, z_2) shows $G(\mathbf{z})$
- one region per digit, morphing at the borders
- empty space between the codes decodes to non-digits

Random Latent Vectors Give No Digits



AE with $d = 16$, decoder applied to random $\mathbf{z}$. Top, $\mathbf{z} \sim \mathcal{N}(0, I)$. Bottom, each z_k drawn with the mean and spread of the codes.

- the codes lie on a thin curved set that a random point misses
- a digit classifier is confident (> 0.9) for 37% of these images and 96% of real digits

Generation needs a known distribution on the latent space.

1

Learning a Probability Distribution



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Generative Models

The data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **samples** of an unknown distribution $p(\mathbf{x})$, handwritten digits or spin configurations at temperature T, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$.

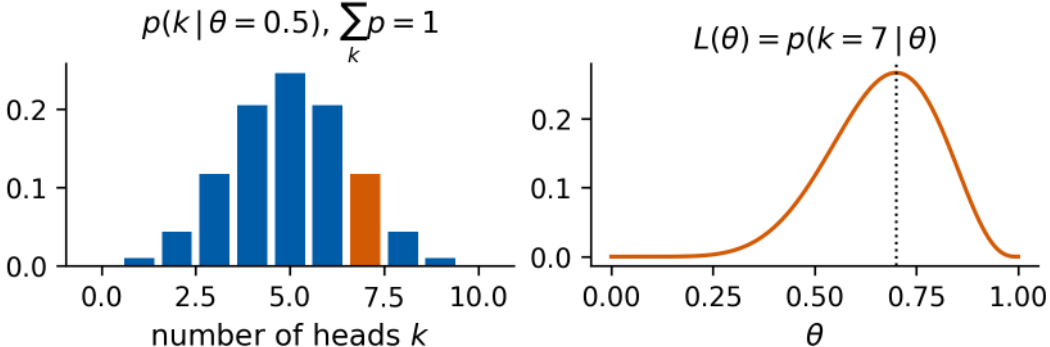
A **generative model**

- fits $p_\theta(\mathbf{x}) \approx p(\mathbf{x})$ to the samples (θ its parameters)
- draws **new samples** that are not in the data set

Uses in science include new molecules and materials, weather ensembles, fast surrogates of simulations and Monte Carlo samplers.

Probability and Likelihood

One model $p(x \mid \theta)$, two readings. **Probability**, θ fixed and x varies, $\sum_x p = 1$. **Likelihood**, data fixed and θ varies, $L(\theta) := p(x_{\text{obs}} \mid \theta)$.

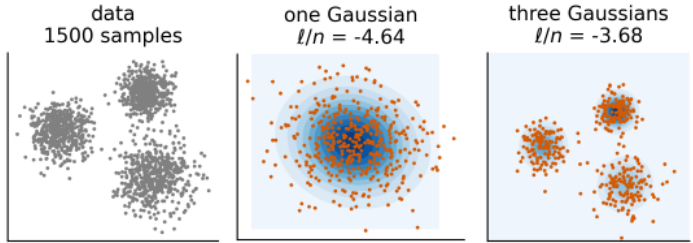


A coin tossed 10 times, $p(k \mid \theta) = \binom{10}{k}\theta^k(1-\theta)^{10-k}$. Left, $p(7 \mid 0.5) = 0.117$. Right, $k = 7$ observed, $\arg\max_\theta L = 0.7$ and $\int_0^1 L\, d\theta = 1/11$, so L is not a distribution over θ.

Maximum Likelihood

$$\theta^* = \arg\max_\theta \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i) \quad \Longleftrightarrow \quad \min_\theta \mathrm{KL}(p \,\|\, p_\theta), \qquad \mathrm{KL}(p\|p_\theta) \coloneqq \mathbb{E}_p \log \frac{p}{p_\theta} \geq 0$$

Kullback–Leibler (KL) divergence, $= 0$ only for $p_\theta = p$ (a one-line proof is in the notes).
Expectation, $\mathbb{E}_p f \coloneqq \int p\, f \approx \frac{1}{n}\sum_i f(\mathbf{x}_i)$, an average over samples.



Mixture of Gaussians $p_\theta = \sum_k \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$, weights π_k, means $\boldsymbol{\mu}_k$, covariances Σ_k, fitted by gradient ascent; orange, 400 new samples; titles, $\ell/n =$ mean $\log p_\theta$ per point.

A Latent Vector Is Like a Macrostate

Two-step generation, $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, I)$, then $\mathbf{x} \sim p_\theta(\mathbf{x} \mid \mathbf{z})$,

$$p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x} \mid \mathbf{z})\, d\mathbf{z}$$

Latent-variable model

$\mathbf{z}$, a few hidden numbers

$\mathbf{x}$, a full image drawn given $\mathbf{z}$

$p_\theta(\mathbf{z}|\mathbf{x})$, which $\mathbf{z}$ produced $\mathbf{x}$

Analogy, statistical mechanics

macrostate, e.g. the magnetisation m

microstate, all N spins at given m

m measured on a snapshot

Prior, Decoder, Evidence and Posterior

$$\underbrace{p_\theta(\mathbf{z}\,|\,\mathbf{x})}_{\text{posterior}} = \frac{\overbrace{p_\theta(\mathbf{x}\,|\,\mathbf{z})}^{\text{decoder}}\ \overbrace{p(\mathbf{z})}^{\text{prior}}}{\underbrace{p_\theta(\mathbf{x})}_{\text{evidence}}} \qquad \text{(Bayes' rule)}$$

- **prior** $p(\mathbf{z}) = \mathcal{N}(0, I)$, fixed before training
- **decoder** $p_\theta(\mathbf{x}|\mathbf{z})$, learned, probability of the image given $\mathbf{z}$
- **evidence** $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$, the model of the data, what maximum likelihood raises
- **posterior** $p_\theta(\mathbf{z}|\mathbf{x})$, the $\mathbf{z}$ that could have produced $\mathbf{x}$

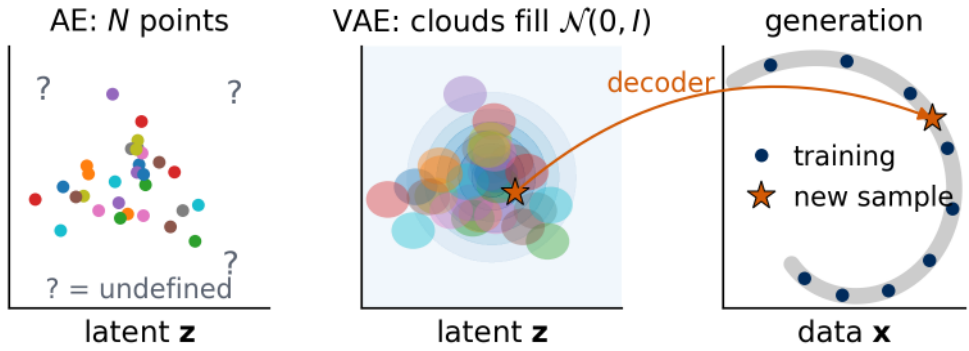
The evidence cannot be computed. A grid of 10 values per coordinate has 10^{16} points for $d = 16$, and random prior draws miss the $\mathbf{z}$ that produce a given digit.

2

Variational Autoencoders

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From Points to a Continuous Distribution



- AE, N images give N latent points with gaps between them
- VAE, one Gaussian cloud $q_\phi(\mathbf{z}|\mathbf{x})$ per image, all pulled onto $\mathcal{N}(0, I)$

The Three Parts of a Variational Autoencoder

- **prior** $p(\mathbf{z}) = \mathcal{N}(0, I_d)$, fixed
- **decoder** $p_\theta(\mathbf{x} \mid \mathbf{z}) = \prod_{j=1}^{784} \tilde{x}_j^{x_j} (1 - \tilde{x}_j)^{1-x_j}$, $\tilde{\mathbf{x}} = \text{sigm}(G_\theta(\mathbf{z}))$, independent Bernoulli pixels
- **encoder** $q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag} \boldsymbol{\sigma}_\phi^2(\mathbf{x}))$, a cloud that approximates the posterior $p_\theta(\mathbf{z}|\mathbf{x})$

$\text{sigm}(u) = 1/(1 + e^{-u})$ is the logistic function, $\tilde{x}_j$ the probability that pixel j is black.

D. P. Kingma, M. Welling, ICLR 2014, arXiv:1312.6114; D. J. Rezende, S. Mohamed, D. Wierstra, ICML 2014, PMLR 32, 1278

Sampling the Prior Misses the Data

Two estimates of $\log p_\theta(\mathbf{x})$ with K samples $\mathbf{z}_k$,

$$\underbrace{\log \frac{1}{K}\sum_{k=1}^{K} p_\theta(\mathbf{x}|\mathbf{z}_k)}_{\mathbf{z}_k \sim \mathcal{N}(0,I)} \qquad \text{and} \qquad \underbrace{\log \frac{1}{K}\sum_{k=1}^{K} \frac{p_\theta(\mathbf{x}|\mathbf{z}_k)\, p(\mathbf{z}_k)}{q_\phi(\mathbf{z}_k|\mathbf{x})}}_{\mathbf{z}_k \sim q_\phi(\mathbf{z}|\mathbf{x}) \text{ (importance sampling)}}$$

- $d = 16$, $K = 1000$, mean over 200 test digits
- prior -154.8 nats, encoder -93.3 nats
- almost no prior draw lands where $p_\theta(\mathbf{x}|\mathbf{z})$ is large

Both estimate the same $\log p_\theta(\mathbf{x})$, in nats (natural logarithm). The prior estimate is far too low because its draws miss the important $\mathbf{z}$.

The encoder tells where to look in the latent space.

The ELBO Follows from Bayes' Rule

Bayes' rule, $p_\theta(\mathbf{x}) = p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z}) / p_\theta(\mathbf{z}|\mathbf{x})$, holds for every $\mathbf{z}$. Take the log,

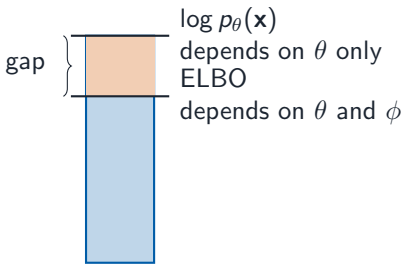
$$\log p_\theta(\mathbf{x}) = \log p_\theta(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}) - \log p_\theta(\mathbf{z}|\mathbf{x})$$

Average both sides over $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$ (the left side does not depend on $\mathbf{z}$), and add and subtract $\log q_\phi$,

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\big[\log p_\theta(\mathbf{x}|\mathbf{z})\big] - \mathrm{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\right)}_{\mathrm{ELBO}(\mathbf{x}),\ \text{computable}} + \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p_\theta(\mathbf{z}|\mathbf{x})\right)}_{\text{gap} \geq 0,\ \text{not computable}}$$

- first term, **reconstruction**: points of the cloud should decode back to $\mathbf{x}$
- second term, **KL to the prior**: the cloud should stay close to $\mathcal{N}(0, I)$
- a KL divergence is never negative, so $\mathrm{ELBO}(\mathbf{x}) \leq \log p_\theta(\mathbf{x})$, the **evidence lower bound**

The Gap Is the Distance to the True Posterior



- raising the ELBO in ϕ cannot move $\log p_\theta(\mathbf{x})$, so it shrinks the gap, **the encoder learns the posterior**
- raising it in θ **improves the decoder**, the model of the data, while the gap stays small
- the gap is zero when the cloud equals the true posterior

The Bound for a Model with Two Latent Values

$z \in \{1, 2\}$, $p(z) = \frac{1}{2}$; decoder $p(\mathbf{x}|z{=}1) = 0.8$, $p(\mathbf{x}|z{=}2) = 0.1$

- evidence $p(\mathbf{x}) = 0.8 \cdot 0.5 + 0.1 \cdot 0.5 = 0.45$, $\log p(\mathbf{x}) = -0.799$; posterior $p(z{=}1|\mathbf{x}) = 0.40/0.45 = 0.889$
- guess $q(z)$ of the posterior, ELBO $= \sum_z q(z)\big[\log p(\mathbf{x}|z) + \log p(z) - \log q(z)\big]$

guess $q(1), q(2)$	reconstruction	KL to prior	ELBO	gap to -0.799
0.5, 0.5 (prior)	-1.263	0.000	-1.263	0.464
0.7, 0.3	-0.847	0.082	-0.929	0.131
0.889, 0.111 (posterior)	-0.454	0.344	-0.799	0

Following the image costs KL to the prior and gains more in reconstruction. The closer the guess to the posterior, the smaller the gap, and at the posterior the bound is exact.

The Loss Is Minus the ELBO

One draw $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon$ per image estimates $\mathbb{E}_{q_\phi}$; μ_k, σ_k, encoder outputs for $\mathbf{x}$,

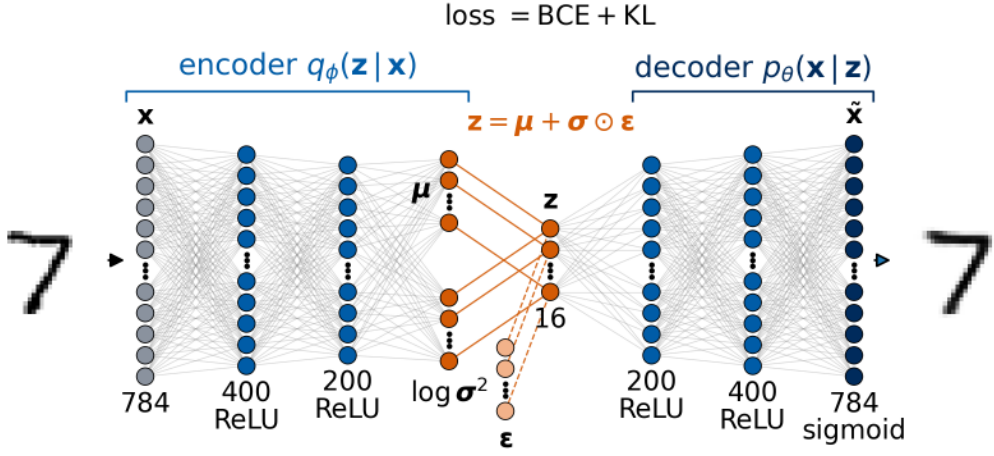
$$\mathcal{L}(\mathbf{x}) = -\,\mathrm{ELBO} = \underbrace{\mathrm{BCE}(\mathbf{x}, \tilde{\mathbf{x}})}_{-\log p_\theta(\mathbf{x}|\mathbf{z})} + \underbrace{\frac{1}{2}\sum_{k=1}^{d} \left(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1\right)}_{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(0,I))}$$

VAE with $d = 16$ on MNIST test digits, nats per image (KL formula derived in the notes)

$\log p_\theta(\mathbf{x})$, wanted	importance sampling, 200 images	-93.3
ELBO, raised by training	same 200 images	-98.1
gap	difference	4.8
loss $= -$ ELBO, printed in training	all test images	101.0
KL part, price of the information in $\mathbf{z}$	all test images	22.7

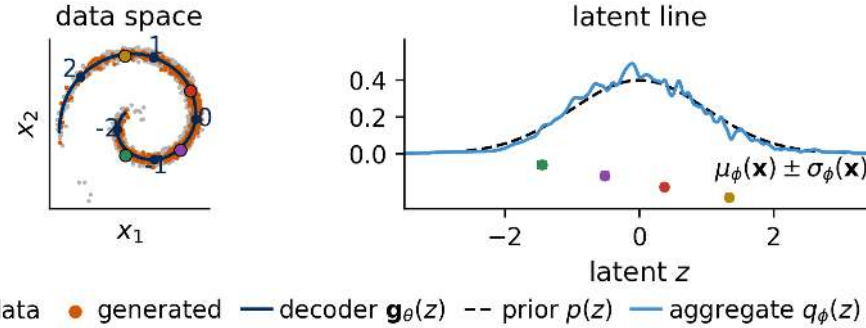
BCE pulls each cloud's points back to $\mathbf{x}$; KL pulls every cloud towards $\mathcal{N}(0, I)$, so a fresh $\mathbf{z}$ lands where the decoder was trained.

The VAE Network Has Two Heads and a Sampling Node



$\varepsilon \sim \mathcal{N}(0, I)$ is a separate input without parameters, and $z_j = \mu_j + \sigma_j \varepsilon_j$ element by element (orange lines), so the gradient reaches the encoder.

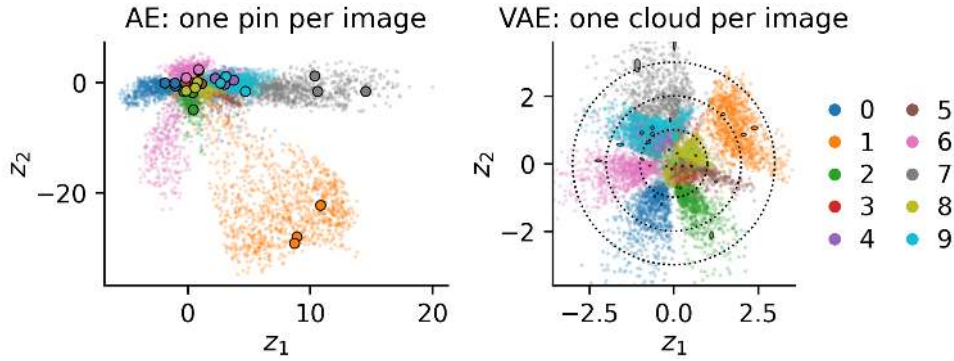
A VAE on Points near a Curve



$\mathbf{x} = \mathbf{f}(z_{\text{true}}) + 0.08\,\boldsymbol{\eta}$ on a spiral, 1D latent space; clouds $\mu_\phi \pm \sigma_\phi$, median $\sigma_\phi = 0.04$.

- **aggregate posterior** $q_\phi(z) \coloneqq \frac{1}{n}\sum_i q_\phi(z|\mathbf{x}_i)$ close to $p(z)$
- $|\mathrm{corr}(\mu_\phi, z_{\text{true}})| = 0.993$

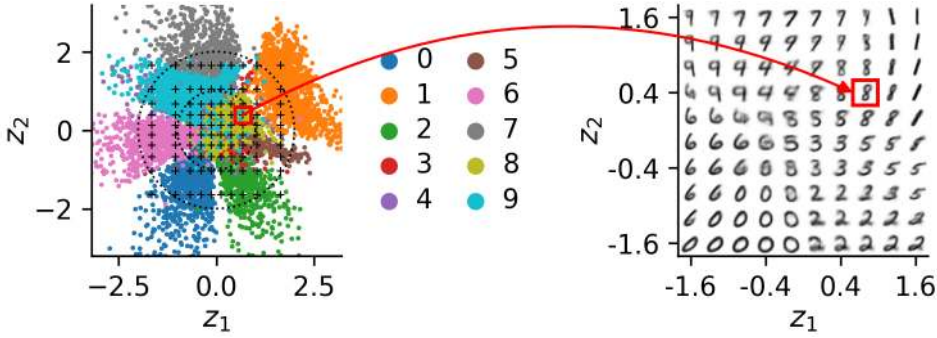
The KL Term Turns Islands into One Covered Disc



$d = 2$; dots, test digits; ellipses $\boldsymbol{\mu}_\phi \pm \boldsymbol{\sigma}_\phi$, three per class; dotted, $|\mathbf{z}| = 1, 2, 3$.

- standard deviation of the codes, AE 3.8, 7.5, VAE 1.03, 1.09

Every Grid Point of the VAE Decodes to a Digit



VAE, $d = 2$; cell, decoder mean $\mathrm{sigm}(G_\theta(\mathbf{z}))$; grid points at equal prior mass $1/100$ per cell.

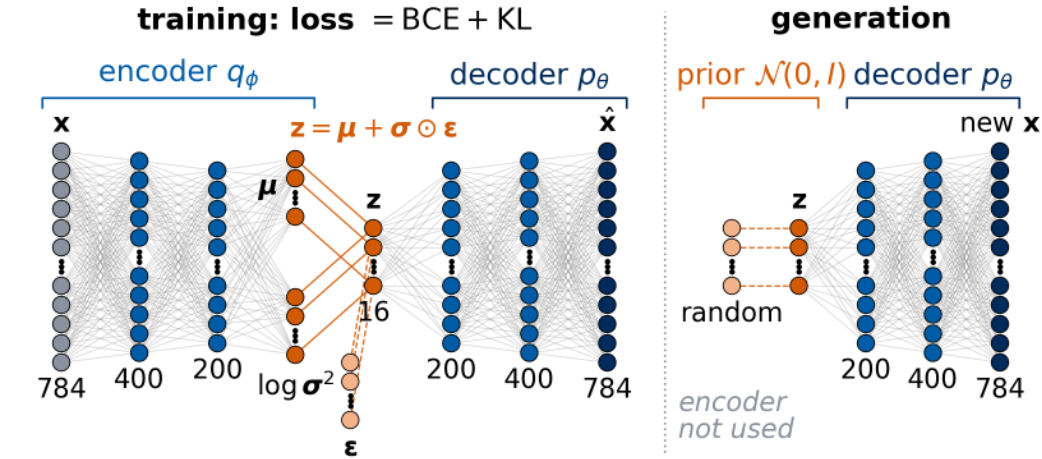
- no nonsense cells, one region per digit
- blurry, since the mean averages all images compatible with $\mathbf{z}$

3

Sampling and Judging a Trained VAE

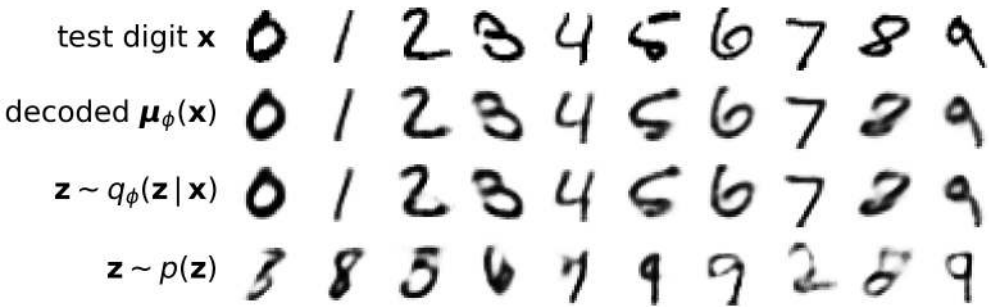
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Training and Generation Use Different Parts of the VAE



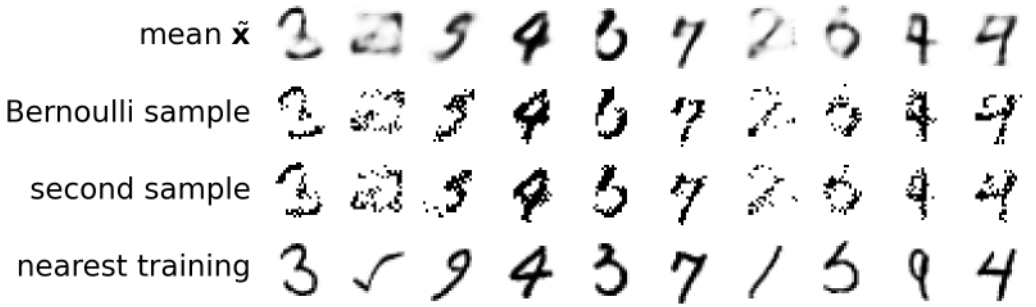
Generation draws **z** from the prior, then **x** from the decoder; the encoder is not used.

Reconstruction and Generation



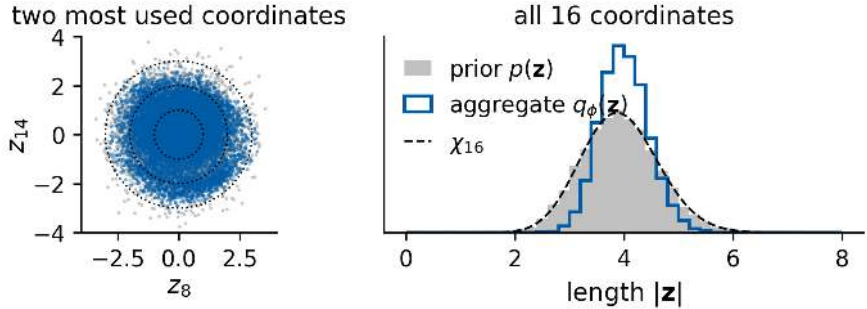
- **reconstruction** $\mathbf{x} \to \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) \to p_\theta(\mathbf{x}|\mathbf{z})$, identity and style kept
- **generation** $\mathbf{z} \sim p(\mathbf{z}) \to p_\theta(\mathbf{x}|\mathbf{z})$
- $d = 16$; classifier confidence > 0.9 for real 96%, reconstructed 92%, generated 54%

A Sample Draws Every Pixel Independently



- decoder mean $\tilde{x}_j$, blurry; sample, pixel j black with probability $\tilde{x}_j$, speckled
- correlations such as a continuous stroke pass through $\mathbf{z}$
- distance to the nearest training image, generated 4.48, unseen test 4.19

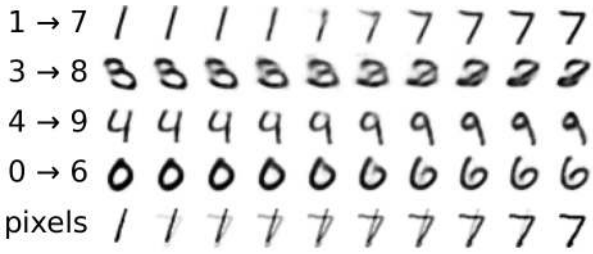
The Encoded Data Do Not Fill the Prior Exactly



Aggregate posterior $q_\phi(\mathbf{z})$ of the test images; grey, $\mathcal{N}(0, I)$; χ_{16}, law of $|\mathbf{z}|$ under the prior.

- per-coordinate standard deviation 0.97 to 1.04
- distance to the nearest test code, prior samples 2.61, codes 1.55, so the prior has **holes**

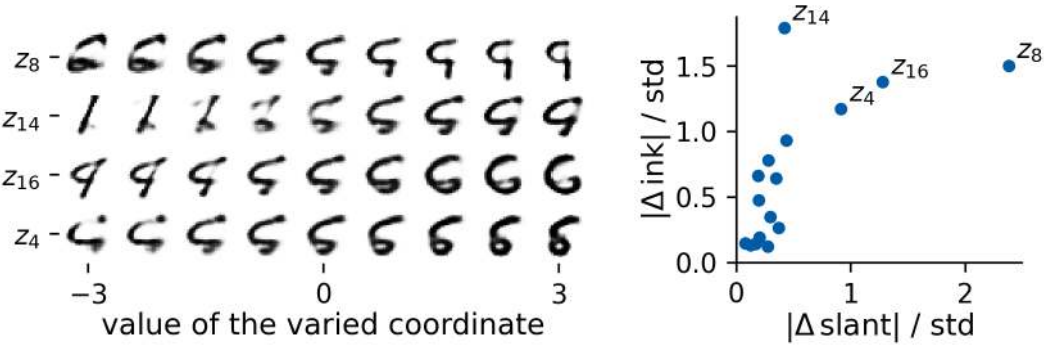
Straight Lines in the Latent Space Stay among the Data



$\mathbf{z}(t) = (1-t)\,\boldsymbol{\mu}_\phi(\mathbf{x}_a) + t\,\boldsymbol{\mu}_\phi(\mathbf{x}_b)$, decoded, $d = 16$; last row, $(1-t)\,\mathbf{x}_a + t\,\mathbf{x}_b$.

- latent line, plausible digits throughout
- pixel line, two superposed digits

One Latent Coordinate Changes Several Properties at Once



Left, one test 5, one coordinate from -3 to 3; the four with the largest $\overline{\mathrm{KL}}_k$, KL term of coordinate k averaged over the data. Right, change of slant and ink for $z_k = -2 \to 2$, in units of their spread over real digits; 7 of 16 coordinates change both by more than 0.3.

A Classifier Trained on Real Data as a Referee

- convolutional neural network (CNN), 3×3 convolutions with 16 and 32 channels, max-pooling, dense 128, 10 outputs
- trained with the cross-entropy on real digits
- applied unchanged to synthetic images

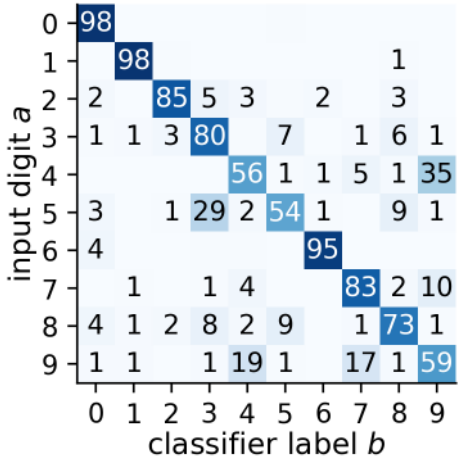
$$p(c \mid \mathbf{x}) = \frac{e^{f_c(\mathbf{x})}}{\sum_{c'} e^{f_{c'}(\mathbf{x})}}$$

test accuracy on 10 000 real digits

0.9854

Every test of synthetic data is first run on real data.

Confusion Matrix of VAE Reconstructions



Row a, digit of the test input; column b, classifier label of its reconstruction; C_{ab} in % of row a; $d = 2$.

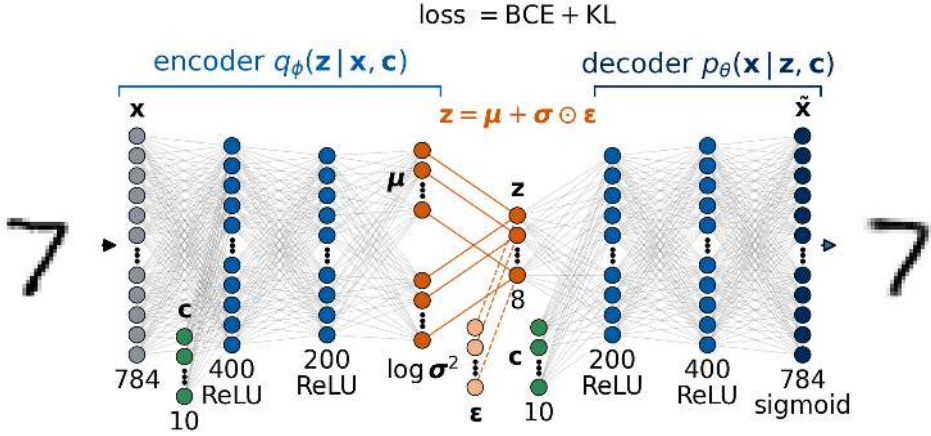
- mean diagonal, real 0.985, $d = 16$ 0.953, $d = 2$ 0.783
- $d = 2$, look-alike digits share a region; 35% of the 4s read as 9

4

Conditional Generation

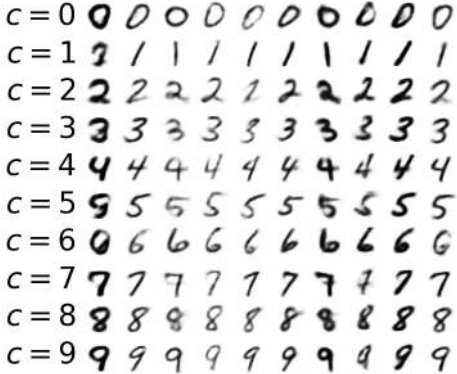
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The Conditional VAE Receives the Label as an Input



Conditional VAE (cVAE), a known factor **c** enters encoder and decoder. Here **c** is the one-hot digit, $c_j = \delta_{j,y}$, and $d = 8$. K. Sohn, H. Lee, X. Yan, NeurIPS 2015, NeurIPS proceedings

Digits on Demand

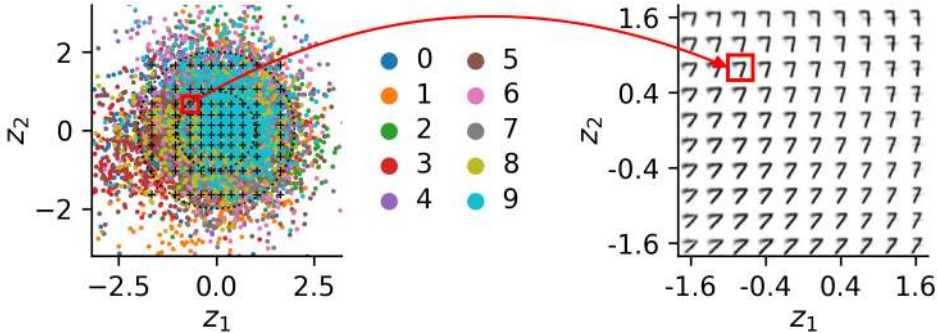


$\mathrm{sigm}(G_\theta(\mathbf{z}, \mathbf{c}))$; row, $\mathbf{c}$ = digit 0 to 9; column, one $\mathbf{z} \sim \mathcal{N}(0, I)$ shared by all rows. The digit changes along a column, the style stays.



Style transfer, $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c}_y)$ of a test digit (left) decoded with $\mathbf{c} = 0, \ldots, 9$.

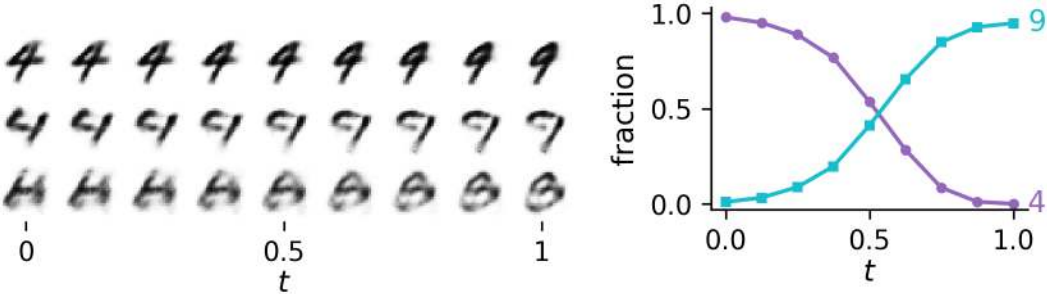
The Latent Plane of a cVAE Holds Only the Style



cVAE, $d = 2$. Left, $\boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c})$ coloured by label. Right, decoder mean with $\mathbf{c} = 7$ at each grid point.

- digit from the nearest class mean in the plane, VAE 0.73, cVAE 0.18 (chance 0.1)
- every cell is a 7 written in a different style

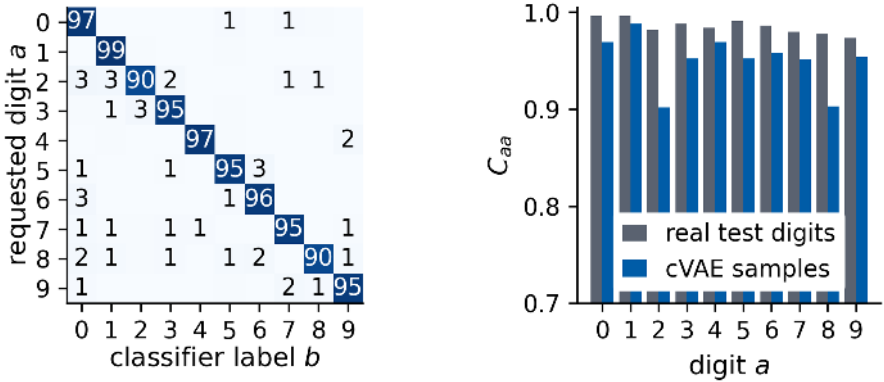
A Condition Never Seen in Training Is Interpolated



$\mathbf{c}(t) = (1-t)\,\mathbf{e}_4 + t\,\mathbf{e}_9$. Left, three fixed $\mathbf{z}$. Right, fraction of 1000 samples assigned to 4 and to 9.

- only $t = 0, 1$ were seen in training
- for a temperature, intermediate values are physical and can be checked against Monte Carlo

The Requested Digit Is Generated



1000 samples per requested digit; C_{ab}, % generated with $\mathbf{c} = a$ and assigned to b; right, C_{aa} and the classifier on real digits. Mean $C_{aa} = 0.950$; largest confusion 3.2% (2 read as 0).

5

Generated Spins and the Hamiltonian



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The Hamiltonian Fixes the Distribution

Spins $s_i = \pm 1$ on an $L \times L$ periodic lattice, $N = L^2$, $k_B = J = J_1 = 1$, $g := J_2/J_1$,

$$H_{\text{Ising}} = -\sum_{\langle ij \rangle} s_i s_j, \qquad H_{J_1 J_2} = -\sum_{\langle ij \rangle} s_i s_j + g \sum_{\langle\langle ij \rangle\rangle} s_i s_j, \qquad p(\mathbf{s} \mid T) = \frac{e^{-H(\mathbf{s})/T}}{Z}$$



J_1, $J_2 = g$, stripes

- Ising ($g = 0$), continuous transition at $T_c = 2/\ln(1+\sqrt{2}) = 2.269$
- J_1–J_2 ground state, ferromagnet ($e = -2 + 2g$) for $g < \frac{1}{2}$, stripes ($e = -2g$) for $g > \frac{1}{2}$
- order parameters $m := \frac{1}{N}\sum_i s_i$; stripes $m_x := \frac{1}{N}\sum_{x,y}(-1)^x s_{x,y}$, m_y likewise, $m_{\text{str}} := (m_x^2 + m_y^2)^{1/2}$

L. Onsager, Phys. Rev. **65**, 117 (1944); D. P. Landau, K. Binder, Phys. Rev. B **31**, 5946 (1985)

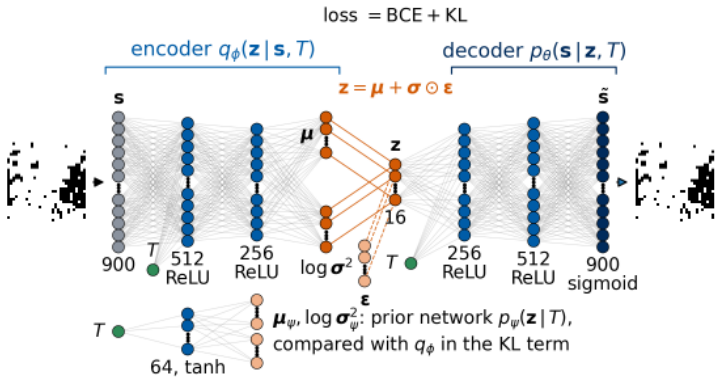
Observables Estimated from Samples

Monte Carlo and generator are judged by the same sample means $\overline{O} = \frac{1}{n}\sum_k O(\mathbf{s}^{(k)})$.

energy per spin	$\overline{e}$, $e_k = H(\mathbf{s}^{(k)})/N$	bonds
magnetisation	$\overline{\lvert m\rvert}$	order
specific heat	$c = (\overline{H^2} - \overline{H}^2)/(NT^2)$	energy fluctuations
susceptibility	$\chi = \frac{N}{T}(\overline{m^2} - \overline{\lvert m\rvert}^2)$	order fluctuations
Binder cumulant	$U_4 = 1 - \overline{m^4}/3\,\overline{m^2}^2$	shape of $P(m)$
correlation	$G(r)$, mean of $s_{\mathbf{x}} s_{\mathbf{x}+r\hat{\mathbf{e}}_\mu}$; $G(1) = -\overline{e}/2$ (Ising)	range

Noise ratio $R_O \coloneqq \mathrm{mean}_T\lvert O_{\mathrm{syn}} - O_A\rvert \,/\, \mathrm{mean}_T\lvert O_B - O_A\rvert$, with A and B two independent Monte Carlo sets; $R_O \approx 1$ is as good as Monte Carlo.

The Spin cVAE with a Prior Network



$\mathbf{c} = T$ (Ising, $L = 30$) or (g, T) (J_1–J_2), $d = 16$. **cVAE**, prior $\mathcal{N}(0, I)$; **+Z2**, trained on $m \geq 0$ representatives, output flipped at random; $+p(\mathbf{z}|\,T)$, prior network.

Given z, Every Generated Spin Is an Independent Coin

One logit $\ell_i(\mathbf{z}, \mathbf{c})$ per site, bits $b_i = (s_i + 1)/2$,

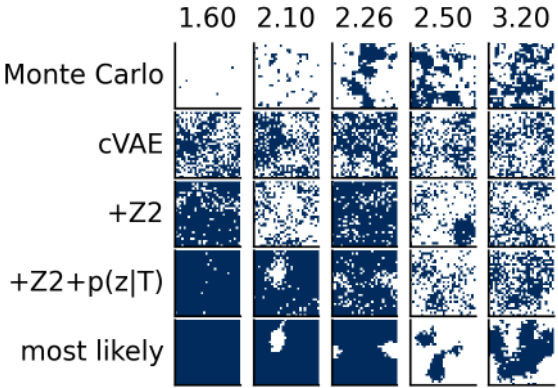
$$p_\theta(\mathbf{s} \mid \mathbf{z}, \mathbf{c}) = \prod_{i=1}^{N} \mathrm{sigm}(\ell_i)^{b_i} \left(1 - \mathrm{sigm}(\ell_i)\right)^{1-b_i}, \qquad \mathrm{Cov}(s_i, s_j) = \mathrm{Cov}_{\mathbf{z}}\left(\tanh \tfrac{\ell_i}{2}, \tanh \tfrac{\ell_j}{2}\right)$$

- a product over sites, the form of a mean-field ansatz
- every correlation passes through 16 latent numbers
- Monte Carlo (Metropolis, Swendsen–Wang) samples $e^{-H/T}/Z$ with all neighbour correlations of H

Independent draws break bonds at random, so e, c and $G(1)$ are the sharp tests.

R. H. Swendsen, J.-S. Wang, PRL **58**, 86 (1987)

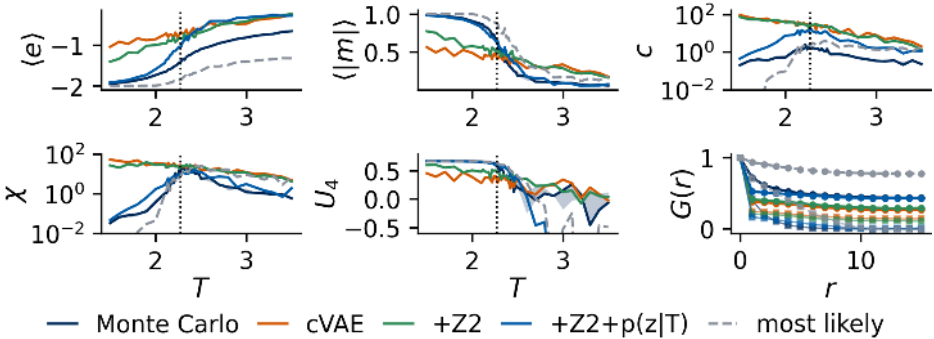
Monte Carlo and Synthetic Ising Configurations



Rows, Monte Carlo; cVAE; $+Z2$; $+p(\mathbf{z}|T)$ with Bernoulli sampling; the same $\mathbf{z}$ with $\tilde{s}_i = \operatorname{sign} \ell_i$ (most likely).

- cVAE and $+Z2$ barely follow T
- $+p(\mathbf{z}|T)$, ordered below T_c, isolated flipped spins
- most likely, large smooth domains even at $T = 3.2$

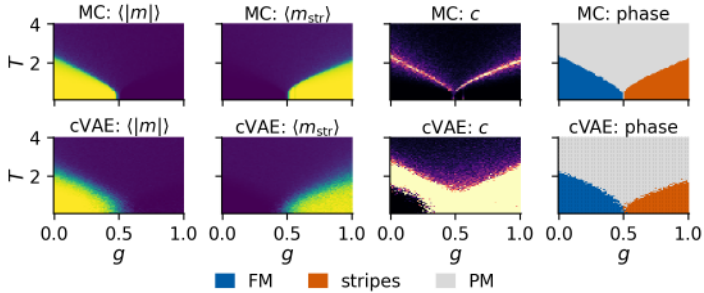
The Order Parameter Is Reproduced and the Energy Fails



$G(r)$ at $T = 2.26$ (circles) and 3.0 (squares); band, Monte Carlo A against B.

- $+p(\mathbf{z}|T)$, $R_{|m|} = 2.8$, $R_\chi = 2.0$, but $R_e = 34.8$, $R_c = 39.0$
- $T = 2.26$, $\overline{e} = -1.43$ (Monte Carlo), -1.04 (Bernoulli), -1.86 (most likely)

The J_1–J_2 Phase Diagram Is Interpolated



Top, Monte Carlo; bottom, cVAE trained at every second g and T; colour 0 to 1 (c to 2.2); phase, CNN vote.

- same phase as Monte Carlo A, set B 99.7 %, cVAE 95.3 % (seen) and 95.4 % (unseen)
- unseen points, $R_{|m|} = 4.8$, $R_{m_{\mathrm{str}}} = 5.9$, $R_e = 63.9$

6

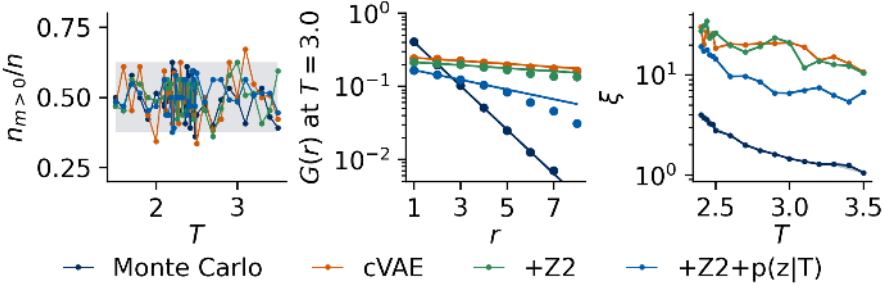
Preserved and Lost Physical Properties



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Spin-Flip Symmetry Kept, Correlation Length Lost

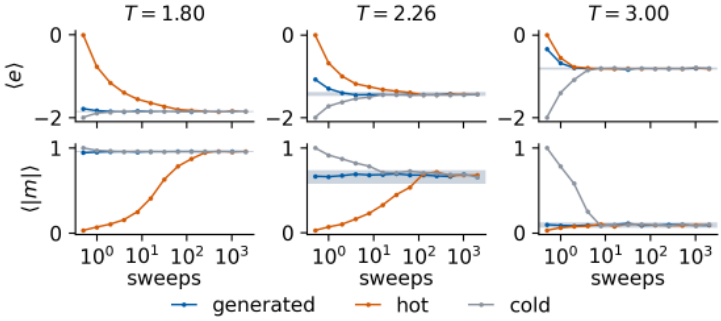
Exact symmetry $P(m) = P(-m)$; above T_c, $G(r) \simeq A\, e^{-r/\xi}$ with correlation length ξ.



- fraction $n_{m>0}/n$ within $\frac{1}{2} \pm 2$ binomial errors (band); no violation at the 5% level ($p \geq 0.07$)
- fits on $r \leq 4$; $T = 3.0$, $\xi = 1.45$ (Monte Carlo), 6.54 ($+p(\mathbf{z}|T)$); A 0.25 against 0.77

Generated Configurations Are Good Monte Carlo Starts

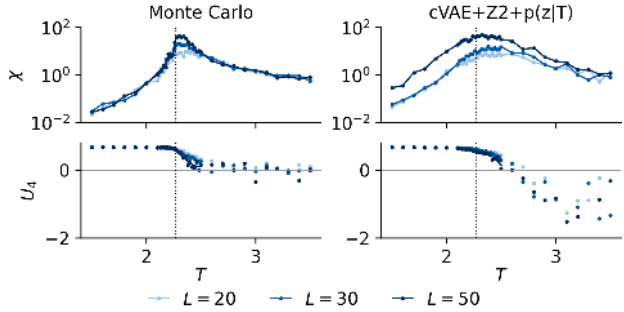
Local Metropolis updates repair bonds in a few sweeps; domain growth is the slow mode.



- 64 chains per start, $L = 30$; band, Monte Carlo mean $\pm$ 3 standard errors
- sweeps to agree in e, $|m|$, $G(r)$ near T_c, generated 2, cold 16, hot 128
- correct statistics come from the Markov chain

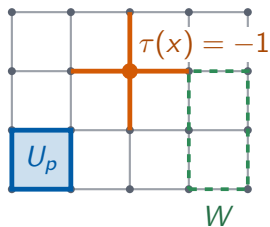
Susceptibility Peak Growth Kept, Binder Crossing Lost

Finite lattice, $\chi_{\max} \propto L^{7/4}$ (2D Ising class); Binder cumulants of all L cross at T_c.



- slope of $\ln \chi_{\max}$ against $\ln L$, Monte Carlo 1.58, generator 1.92
- U_4 of $L = 30$, 50 cross at 2.252 (Monte Carlo); synthetic, none near T_c
- synthetic $U_4 < 0$ above T_c, a $P(m)$ with heavy tails

The Ising Lattice Gauge Theory Has a Local Symmetry



links $\sigma_\mu(x) = \pm 1$, cubic lattice, $L = 6$, 648 links; plaquette U_p, product of the four links of a square,

$$H = -\sum_p U_p, \qquad \beta_g := 1/T$$

$\sigma_\mu(x) \to \tau(x)\,\sigma_\mu(x)\,\tau(x + \hat{\mathbf{e}}_\mu)$, $\tau(x) = \pm 1$, leaves every U_p unchanged

- Elitzur, a local symmetry cannot break, so $\langle \sigma_\mu(x) \rangle = 0$ at every β_g
- gauge-invariant tests, $\langle U_p \rangle$, c, Wilson loops $W(R, S)$ (product of links around an $R \times S$ rectangle)

F. J. Wegner, J. Math. Phys. **12**, 2259 (1971); S. Elitzur, Phys. Rev. D **12**, 3978 (1975)

Wilson Loops Distinguish Confinement from Deconfinement

Transition at $\beta_g = 0.7614$.

Confined, $\beta_g < 0.7614$

area law $W(R,S) \sim e^{-\sigma_s RS}$, string tension $\sigma_s > 0$

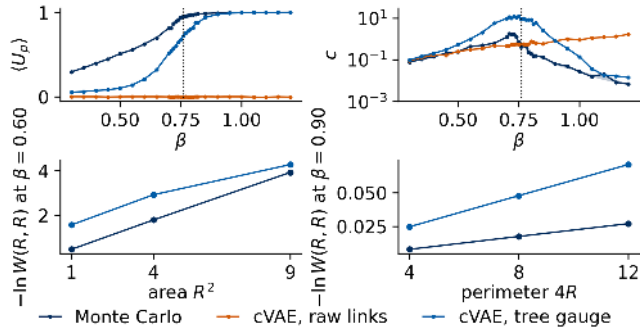
Deconfined, $\beta_g > 0.7614$

perimeter law, $-\ln W(R,S) \propto 2(R+S)$

- Creutz ratio $\chi_C(2,2) = -\ln\left[W(2,2)\,W(1,1)/W(2,1)^2\right]$ estimates σ_s
- stored links are randomly gauge transformed, every link ± 1 with probability $\frac{1}{2}$
- tree gauge, links of a maximal tree set to $+1$, one configuration per gauge orbit (as $m \geq 0$ for spins)

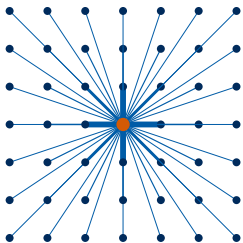
M. Creutz, Phys. Rev. D **21**, 2308 (1980)

Gauge Theory, Perimeter Law Kept, String Tension Lost



- axis $\beta = \beta_{\mathrm{g}}$; raw links, $-\,\mathrm{ELBO} = 449.1 \approx N \ln 2 = 449.2$ nats, nothing learned
- tree gauge, deconfined, $\langle U_p \rangle = 0.996$ against 0.997 at $\beta_{\mathrm{g}} = 1.00$
- confined, perimeter law; $\chi_{\mathrm{C}}(2,2) = 0.222$ against 0.406 at $\beta_{\mathrm{g}} = 0.60$

Long-Range Couplings Bring the Mean-Field Regime



line width $\propto J_{ij}$, $\alpha = 2$

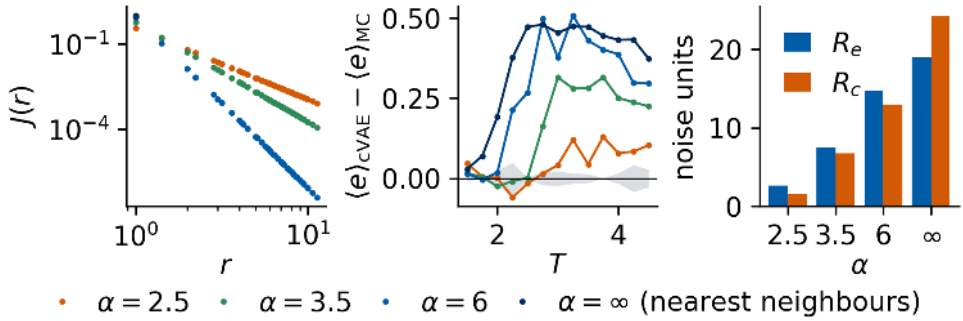
$$H = -\frac{1}{2}\sum_{i\neq j} J_{ij}\, s_i s_j, \qquad J_{ij} = \frac{J_0}{r_{ij}^{\alpha}}, \qquad \sum_j J_{ij} = 4$$

- $\alpha < 3$, mean-field critical exponents in 2D
- $\alpha > 3.75$, short-range Ising exponents; $\alpha = \infty$, nearest neighbours
- the local field averages many spins and fluctuates little

Prediction, the energy failure of a product decoder shrinks as the range grows.

M. E. Fisher, S.-k. Ma, B. G. Nickel, PRL **29**, 917 (1972); J. Sak, Phys. Rev. B **8**, 281 (1973)

The Energy Failure Shrinks with the Interaction Range



$L = 16$, 13 temperatures; left, couplings $J(r)$; middle, energy error, band from two Monte Carlo sets.

- $R_e = 18.9$, 14.8, 7.5, 2.7 for $\alpha = \infty$, 6, 3.5, 2.5; R_c from 24.2 to 1.6
- order parameter reproduced for every α, $R \leq 2.2$; one lattice size, one seed per α

Collective Properties Survive and Local Ones Are Lost

preserved	**lost**
$\langle \lvert m \rvert \rangle$, χ, $P(m)$; $R_{\lvert m \rvert} = 2.8$, $R_\chi = 2.0$	$\langle e \rangle$, c, $P(e)$; $R_e = 34.8$, $R_c = 39.0$
phase at unseen (g, T), 95.4 %	energy at unseen (g, T), $R_e = 63.9$
spin-flip symmetry, $p \geq 0.07$	correlation length, $\xi = 6.54$ against 1.45
growth of $\chi_{\max}$ with L	Binder crossing of $L = 30$ and 50
deconfined plaquettes, perimeter law	confined plaquettes, string tension
energy at long range, $R_e = 2.7$ ($\alpha = 2.5$)	energy at short range, $R_e = 18.9$

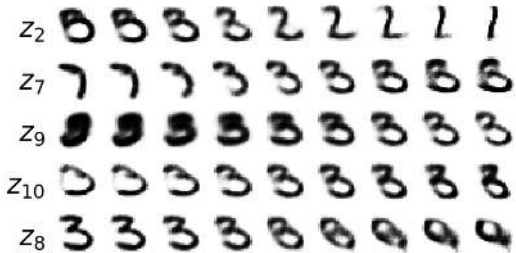
Synthetic configurations answer questions about the phase and give good Monte Carlo starts; they are not samples of $e^{-H/T}/Z$.

7

Disentangled Representations and the β-VAE

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In a Plain VAE One Coordinate Changes Several Features



VAE, $d = 10$, one test 3; row, one z_k from -3 to 3, others fixed

Disentangled code, each generative factor carried by its own coordinate.

- z_2, $3 \to 2 \to 1$ with slant
- z_7, $7 \to 3 \to 8$
- only z_9 changes one property, the thickness

The β-VAE Puts a Price on Information

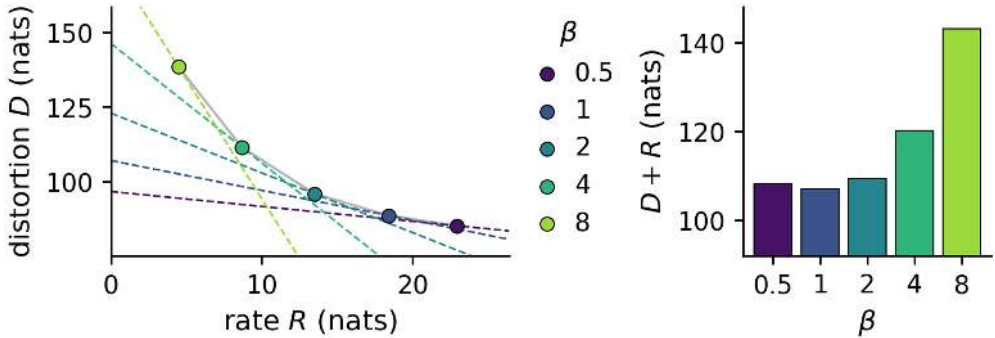
$$\mathcal{L}_\beta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[-\ln p_\theta(\mathbf{x}|\mathbf{z})\big]}_{\text{reconstruction}} + \beta \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\right)}_{\text{KL term}}, \qquad \mathcal{L}_1 = -\,\mathrm{ELBO}$$

- $\beta > 1$, fewer used coordinates, blurrier images, stronger pull to the factorised prior $\prod_k \mathcal{N}(0,1)$
- β is a Lagrange multiplier, best reconstruction under a budget $\mathrm{KL} < \epsilon$
- β is the KL weight throughout; temperatures are written $1/T$

Averaged over the data, distortion $\mathcal{D}$ and rate $\mathcal{R}$ (nats per image), $\mathcal{L}_\beta = \mathcal{D} + \beta\mathcal{R}$.

I. Higgins et al., ICLR 2017, OpenReview; A. A. Alemi et al., ICML 2018, PMLR 80, 159

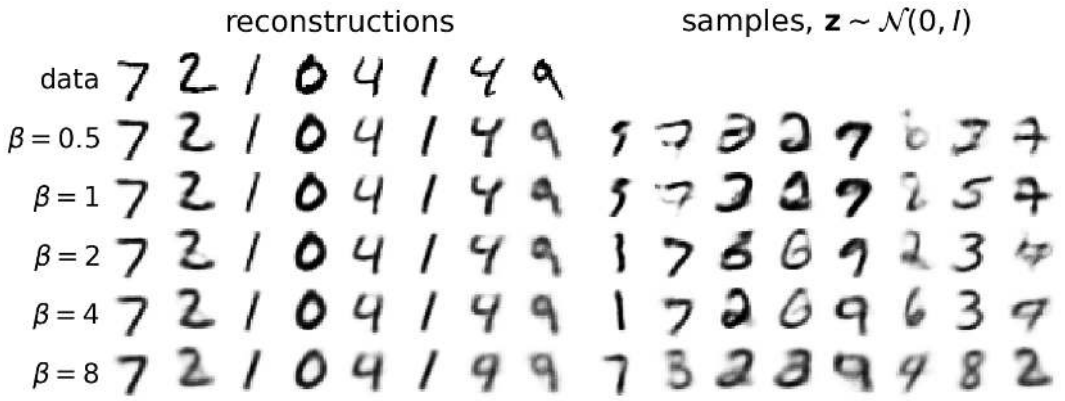
β Is Minus the Slope of the Rate–Distortion Curve



MNIST, $d = 10$; dashed, lines $\mathcal{D} + \beta\mathcal{R} = \text{const}$; right, $-\text{ELBO} = \mathcal{D} + \mathcal{R}$.

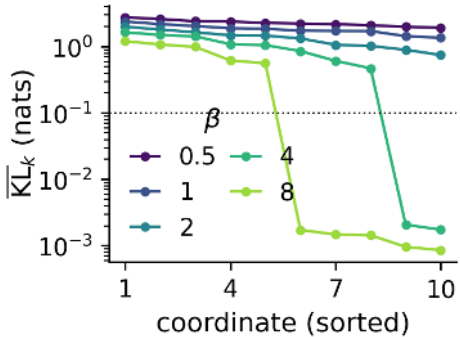
- rate 22.9 $\to$ 4.5 nats, distortion 85.2 $\to$ 138.5 nats for $\beta = 0.5 \to 8$
- $\beta = 0.5$ and 2 within 1.1 and 2.4 nats of the best ELBO, with very different codes

Larger β Blurs Reconstructions and Samples

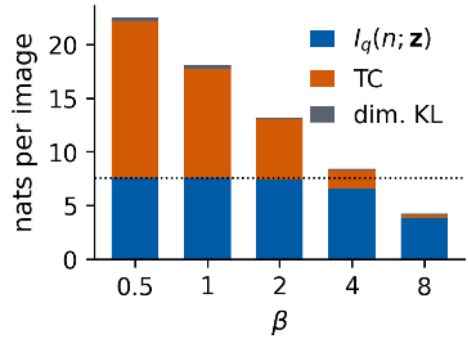


Left, test digits and $\text{sigm}(G_\theta(\mu_\phi(\mathbf{x})))$. Right, $\text{sigm}(G_\theta(\mathbf{z}))$, the same 8 vectors $\mathbf{z} \sim \mathcal{N}(0, I)$ for every β. At $\beta = 8$ detail is lost and the 4 becomes a 9.

Larger β Removes Coordinates and Total Correlation



$\overline{\mathrm{KL}}_k$ sorted; active above 0.1 nats

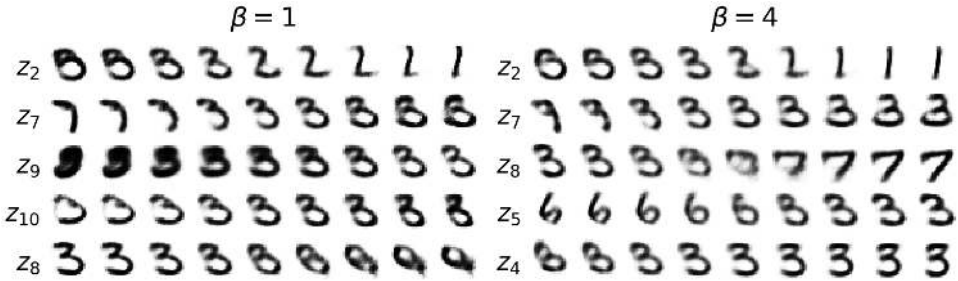


parts of $\mathcal{R}$ on $M = 2000$ digits; dotted, $\ln M$

$\mathcal{R} = I_q$ (index–code information) + total correlation (TC, $= 0$ for independent coordinates) + dimension-wise KL.

- active coordinates 10, 8, 5 of 10 for $\beta = 1, 4, 8$; TC 14.50 $\rightarrow$ 0.26 nats

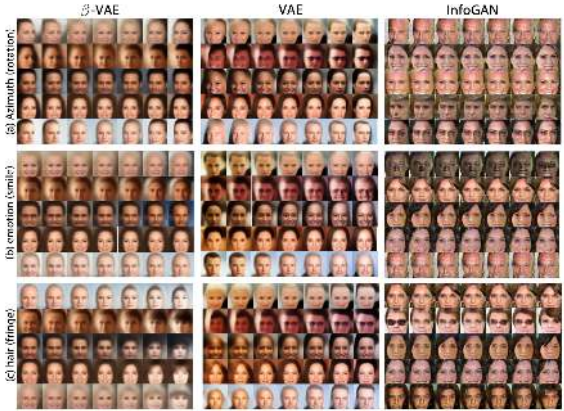
On MNIST, Traversals Change the Digit Identity



Same test 3; row, one active z_k from -3 to 3.

- both models change the identity along every axis; $\beta = 4$ has fewer, somewhat simpler axes
- the identity is discrete and $\mathbf{z}$ continuous, so digits morph
- the true factors of handwriting are unknown

A β-VAE Turns Heads Where a Plain VAE Mixes Factors

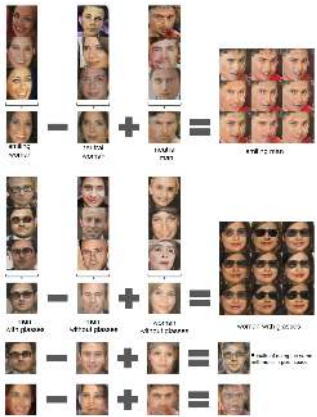


CelebA, 64×64; block, one coordinate from -3 to 3; row, one seed face

- β-VAE ($\beta = 250$), azimuth, smile, fringe, each in one coordinate
- VAE, azimuth mixed with emotion, glasses, gender
- InfoGAN, a GAN-based method

Fig. 1 of I. Higgins et al., ICLR 2017, OpenReview; CelebA, Z. Liu, P. Luo, X. Wang, X. Tang, ICCV 2015, doi:10.1109/ICCV.2015.425

Latent Arithmetic on Faces



- column, latent vectors of 3 faces, averaged
- $\bar{\mathbf{z}}_{\text{man, glasses}} - \bar{\mathbf{z}}_{\text{man}} + \bar{\mathbf{z}}_{\text{woman}}$ decodes to a woman with glasses
- a generative adversarial network (GAN) on web faces; pixel arithmetic (bottom) gives misaligned overlaps

Figure 7 of A. Radford, L. Metz, S. Chintala, arXiv:1511.06434

Small Faces Keep the Global Appearance



top, test faces $\mathbf{x}$; bottom, $\text{sigm}(G_\theta(\boldsymbol{\mu}_\phi(\mathbf{x})))$

32×32 CelebA, $d = 32$, $\beta = 3$.

- rate 15.8 nats per face; 18 of 32 coordinates active
- kept, pose, skin and hair colour, background, shape of the face
- lost, eyes, mouth, glasses

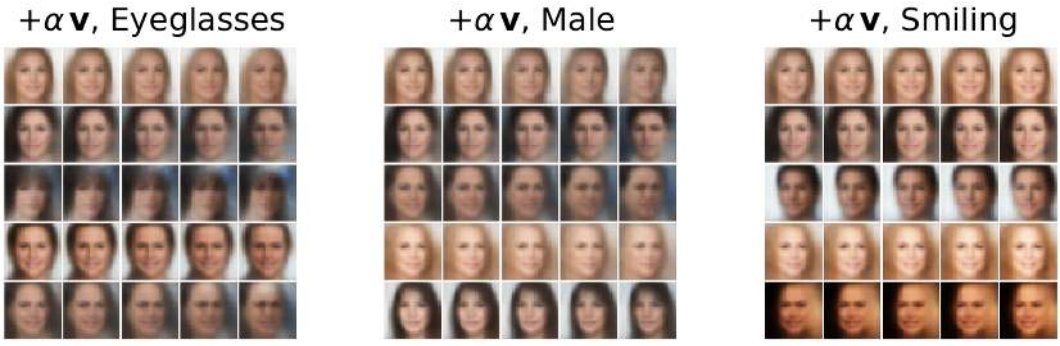
Our Faces Model Separates Background, Light, Skin and Hair



Two test faces; row, one of the 8 coordinates with the largest $\overline{\mathrm{KL}}_k$ from -3 to 3; label, attribute with the largest Pearson $|\rho|$ with μ_k.

- z_{10} background, z_{32} direction of the light, z_{25} skin and hair colour, z_2 hair length
- no coordinate turns the head cleanly; $|\rho| \leq 0.35$ with the 40 labels

Attribute Directions Exist but Are Faint at 32×32



$\mathbf{v}_a = \langle \boldsymbol{\mu}_\phi(\mathbf{x}) \rangle_{a=1} - \langle \boldsymbol{\mu}_\phi(\mathbf{x}) \rangle_{a=0}$, labels used only after training; rows, test faces without a decoded at $\boldsymbol{\mu}_\phi(\mathbf{x}) + \alpha\, \mathbf{v}_a$. Glasses appear as a dark band at the eyes for $\alpha = 2$.

8

Order Parameters from a β-VAE

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Probing the Latent Coordinates with Observables

The network sees only snapshots $\mathbf{s}$, no T, no phase, no Hamiltonian.

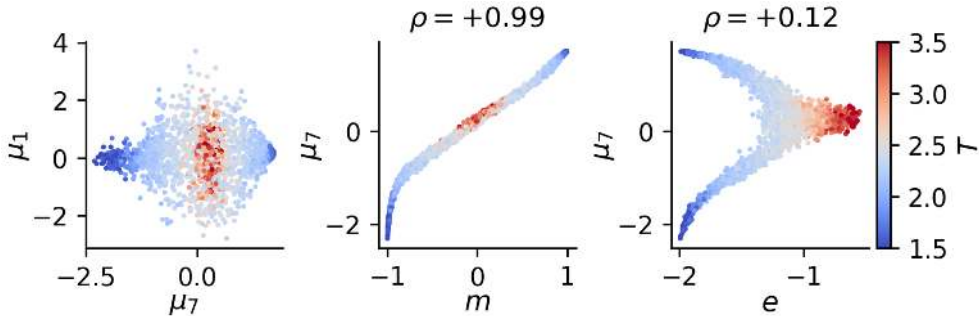
1. β-VAE with Bernoulli decoder, $d = 8$, $\beta \in \{1, 4\}$
2. active coordinates, $\overline{\mathrm{KL}}_k > 0.1$ nats on held-out snapshots
3. Pearson correlation of the latent mean $\mu_k(\mathbf{s})$ with an observable $O(\mathbf{s})$,

$$\rho_{kO} = \frac{\langle \mu_k O \rangle - \langle \mu_k \rangle \langle O \rangle}{\sqrt{\mathrm{Var}\, \mu_k \;\; \mathrm{Var}\, O}} \in [-1, 1]$$

Linear only, $\mu_k \propto m$ gives $|\rho| \approx 1$ with m and $\rho \approx 0$ with $|m|$.

L. Wang, Phys. Rev. B **94**, 195105 (2016); S. J. Wetzel, Phys. Rev. E **96**, 022140 (2017)

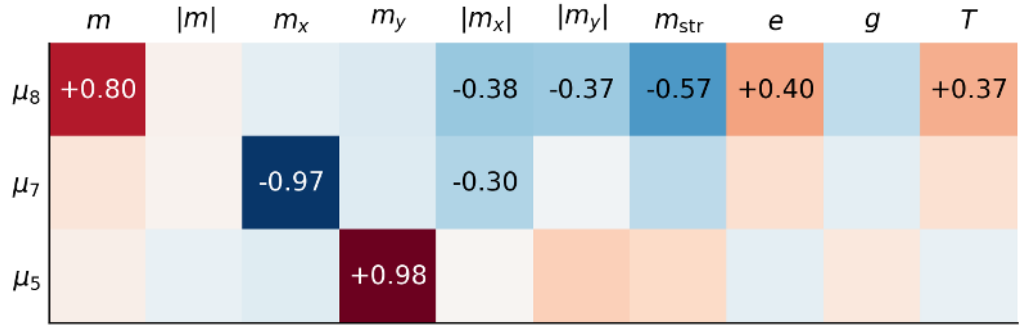
One Coordinate Is the Magnetisation



2D Ising, $L = 30$, $T \in [1.5, 3.5]$, held-out snapshots coloured by T; $\beta = 4$.

- $\rho(\mu_7, m) = +0.99$ ($\beta = 1$ and 4); every other coordinate $|\rho| \leq 0.16$
- ordered phases at the two ends of the μ_7 axis, disordered phase at its centre
- energy follows $|\mu_7|$, $\rho(|\mu_7|, e) = -0.88$

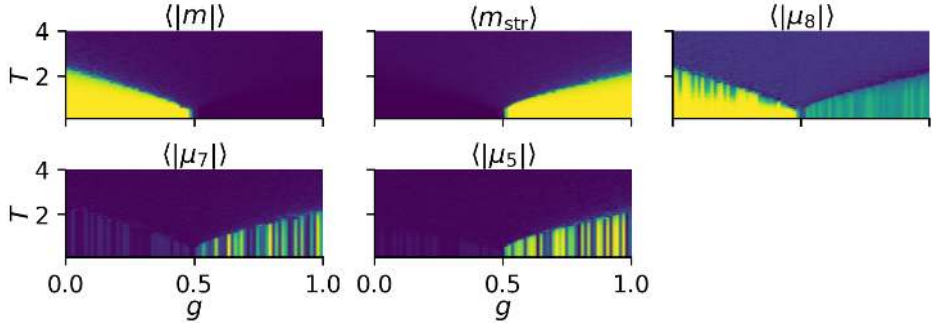
Three J_1–J_2 Coordinates Are m, m_x and m_y



$L = 32$, 81×79 grid of (g, T), 10 training and 2 held-out snapshots per point; $\beta = 4$; numbers if $|\rho| \geq 0.3$.

- $\rho(\mu_7, m_x) = -0.97$, $\rho(\mu_5, m_y) = +0.98$, $\rho(\mu_8, m) = +0.80$
- $\beta = 1$, m_x and m_y mixed, e.g. μ_5 with -0.58 and $+0.67$

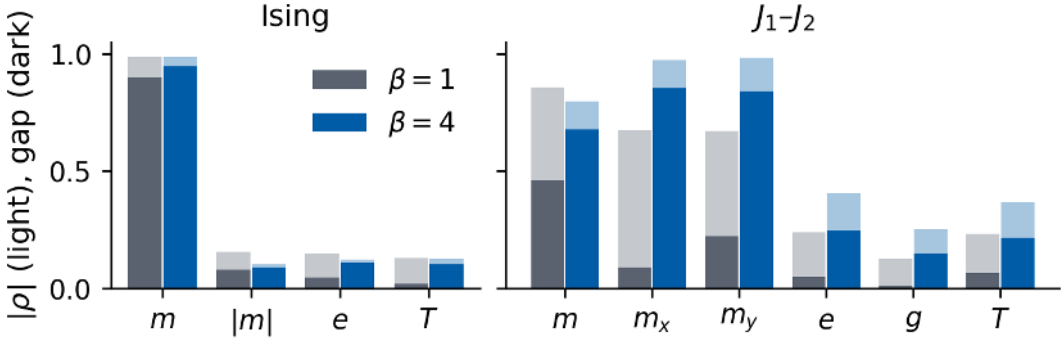
The Coordinates Map the Phase Diagram



Mean of $|m|$, m_{str}, $|\mu_k|$ at each (g, T), $\beta = 4$; colour, minimum (dark) to maximum (yellow).

- $|\mu_8|$, ferromagnet; $|\mu_7|$, $|\mu_5|$, stripe phase
- complementary stripe bands come from the cooled chains of the data

Larger β Gives Each Order Parameter Its Own Coordinate



Light bar, $\max_k |\rho_{kO}|$; dark bar, gap to the second largest, large when only one coordinate follows O.

- Ising m, gap 0.90 ($\beta = 1$) and 0.95 ($\beta = 4$)
- J_1–J_2, m_x 0.09 $\to$ 0.85, m_y 0.22 $\to$ 0.84, m 0.46 $\to$ 0.68

A Coordinate Becomes a Candidate Formula

Least-squares fit of μ_k by 8 standardised features, m, m_x, m_y, their moduli, nearest- and next-nearest-neighbour correlations,

$$R^2 = 1 - \frac{\sum_{\mathbf{s}} (\mu_k - \tilde{\mu}_k)^2}{\sum_{\mathbf{s}} (\mu_k - \langle \mu_k \rangle)^2}$$

- $\beta = 4$, $R^2 = 0.982$ (z_8, mainly m), 0.992 (z_7, m_x), 0.989 (z_5, m_y)
- a low R^2 points to a quantity outside the features
- correlation confirms and does not explain; sign, scale and the found axes depend on model, data, β and seed

F. Locatello et al., ICML 2019, arXiv:1811.12359

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Physical Parameters from Trajectories



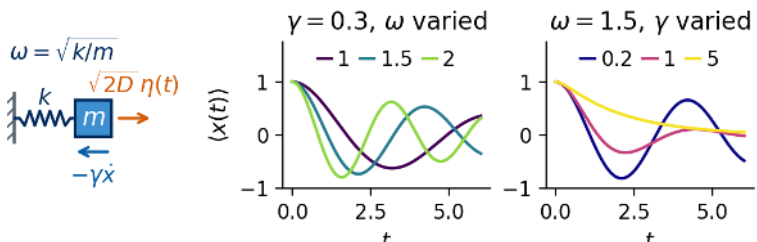
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Frequency and Damping Shape a Noisy Oscillation

$$\ddot{x} = -\omega^2 x - \gamma\,\dot{x} + \sqrt{2D}\,\eta(t), \qquad \langle \eta(t) \rangle = 0, \quad \langle \eta(t)\,\eta(t') \rangle = \delta(t - t')$$

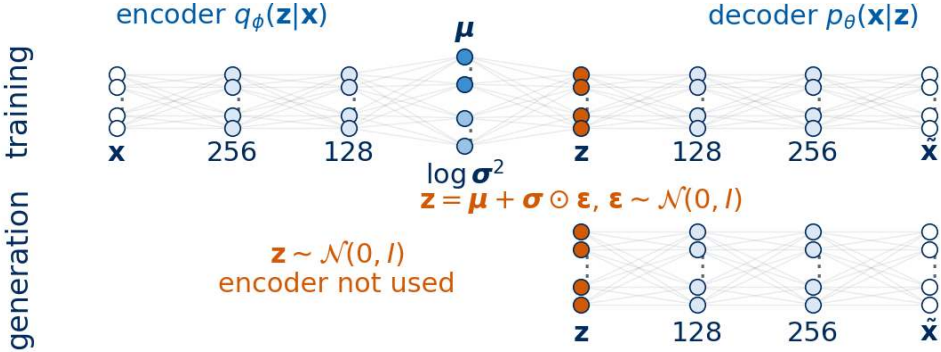
ω, frequency without friction; γ, friction rate; $\eta(t)$, Gaussian white noise; D, noise strength.

- underdamped, $\gamma < 2\omega$, oscillation at $\omega_d = \sqrt{\omega^2 - \gamma^2/4}$ inside $e^{-\gamma t/2}$
- overdamped, $\gamma > 2\omega$, slow creep at the rate $\approx \omega^2/\gamma$

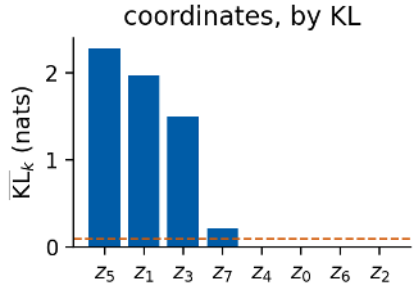


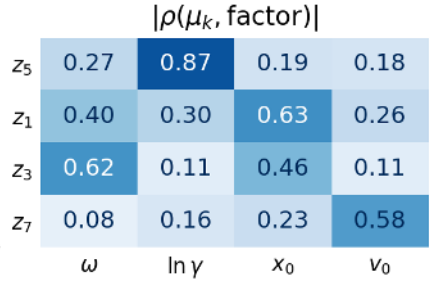
A β-VAE Compresses Each Trajectory

Gaussian decoder of width $\sigma_x = 0.1$, so the reconstruction term is $\|\mathbf{x} - G_\theta(\mathbf{z})\|^2/(2\sigma_x^2)$; $d = 8$, $\beta \in \{0.5, 1, 4, 8\}$, 3 seeds each; factors ω, $\ln\gamma$, x_0, v_0.



Damping Gets Its Own Coordinate and Frequency Is Shared

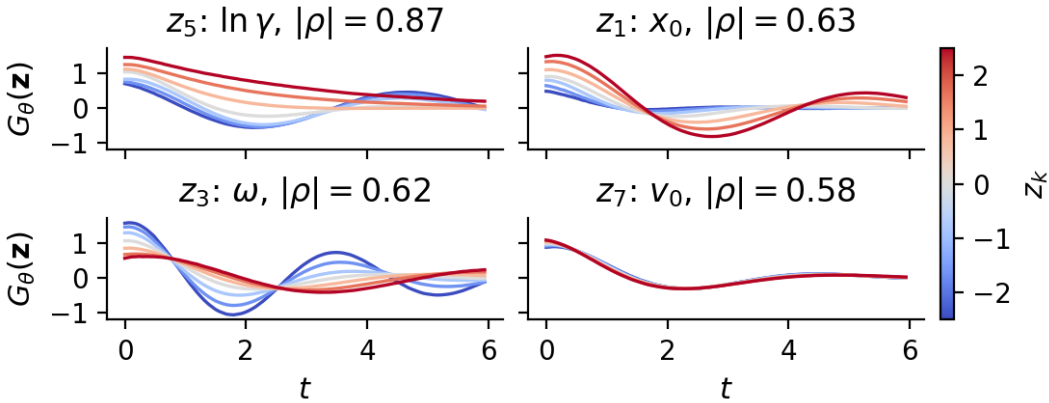




$\beta = 4$, best of three seeds, 5000 test trajectories; $|\rho|$, Pearson correlation of μ_k and factor.

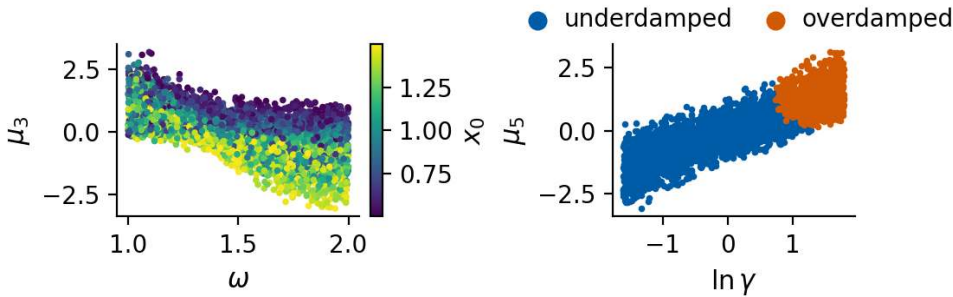
- 4 active coordinates for 4 factors; damping $|\rho| = 0.87$
- ω (0.62) and x_0 (0.63) shared by two coordinates
- mutual information gap (MIG) 0.03 to 0.07 for every β, true factors 0.92

Traversals Separate Damping from Frequency



$\boldsymbol{\mu}$ of one test trajectory ($\omega \approx 1.5$, $\gamma \approx 1$, $x_0 \approx 1$, $v_0 \approx 0$); one z_k from -2.5 to 2.5, the others fixed; title, factor with the largest $|\rho|$. The ω coordinate also moves $x(0)$.

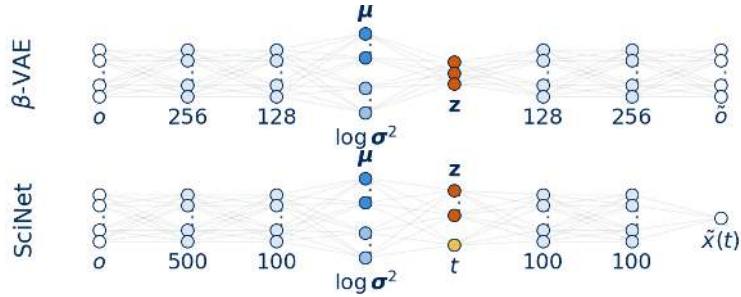
Overdamped Trajectories Reveal Only ω^2/γ



Left, frequency coordinate against ω, colour x_0; right, damping coordinate against $\ln\gamma$.

- $|\rho(\mu_\gamma, \ln\gamma)| = 0.80$ underdamped, 0.13 overdamped
- overdamped, $|\rho(\mu_\gamma, \ln(\omega^2/\gamma))| = 0.69$, the slow rate that the data show
- R^2 of the ω coordinate, 0.38 from ω, 0.63 from $\omega, x_0, \omega x_0$

SciNet Answers Questions about the System



Pendulum without noise, $\ddot{x} = -\kappa x - b\dot{x}$; the decoder predicts $x(t)$ at a requested t.

- cost, prediction error $+\,\beta$ KL with $\beta = 10^{-3}$
- 1000 epochs, $|\rho| = 1.00$ (κ) and 0.98 (b) in separate neurons
- β-VAE with $\beta = 4$, 0.99 and 0.97 after 60 epochs

R. Iten et al., Phys. Rev. Lett. **124**, 010508 (2020), doi:10.1103/PhysRevLett.124.010508

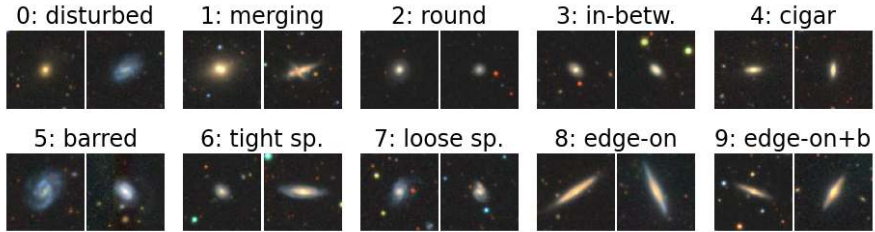
10

Galaxy Morphology with a β-VAE



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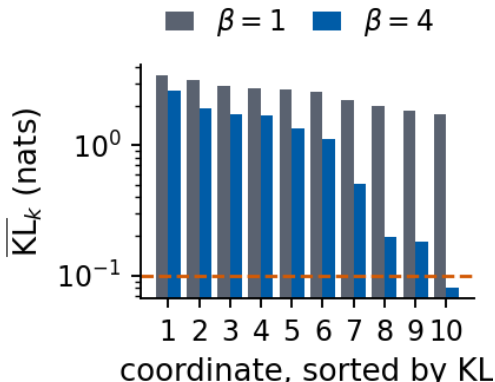
Galaxy10 DECaLS Holds 17 736 Galaxies in Ten Classes



- 64×64 pixels of $1.05''$; bands z (near infrared), r (red), g (blue-green) shown as red, green, blue
- images from the Dark Energy Camera Legacy Survey (DECaLS); classes from Galaxy Zoo votes, not used in training

H. W. Leung, J. Bovy, Galaxy10 DECaLS, Zenodo; M. Walmsley et al., MNRAS 509, 3966 (2022); A. Dey et al., AJ 157, 168 (2019)

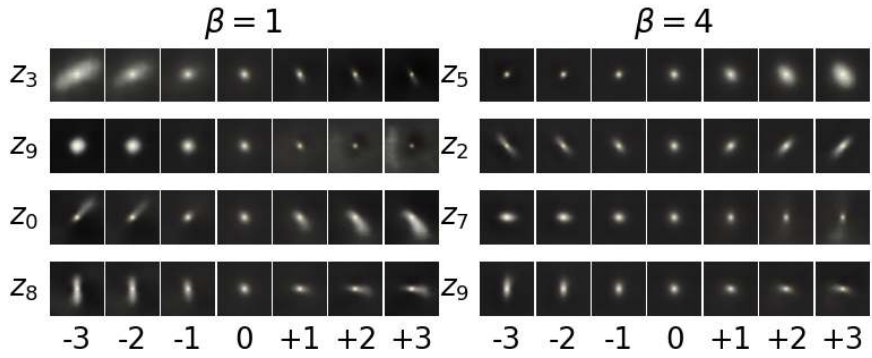
A Larger β Lowers the Rate at Almost the Same Error



Convolutional β-VAE, Gaussian decoder, $d = 10$, 15 000 training images.

- rate 25.3 $\rightarrow$ 11.4 nats for $\beta = 1 \rightarrow 4$
- mean squared error per pixel $3.01 \rightarrow 3.13 \times 10^{-3}$
- active coordinates 10 $\rightarrow$ 9 of 10

Traversals Separate Size, Brightness and Shape



Row, decoded $z_k = \mu_k(\mathbf{x}_0) + v$ for one round smooth galaxy, other coordinates fixed.

- $\beta = 4$, z_5 size and brightness; z_2 diagonal stretch; z_7, z_9 stretch along the axes
- $\beta = 1$, size, brightness and shape change together; no row changes the colour

Latent Coordinates Follow Size, Brightness and Ellipticity

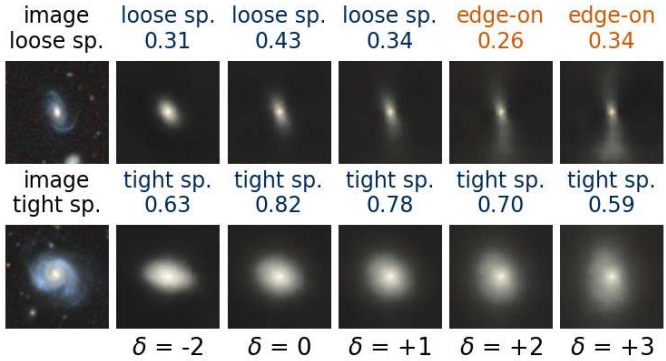
	R	q	e_1	e_2	F	c	C
z_5	0.93	0.06	0.04	0.14	0.97	0.22	0.15
z_2 (e_2)	0.36	0.07	0.06	0.78	0.39	0.06	0.12
z_7 (e_1)	0.14	0.13	0.60	0.05	0.20	0.13	0.14
z_9 (e_1)	0.19	0.12	0.80	0.14	0.21	0.13	0.12
z_3	0.17	0.12	0.48	0.14	0.12	0.11	0.15
z_1	0.19	0.04	0.30	0.33	0.17	0.00	0.02
z_8	0.26	0.05	0.49	0.36	0.26	0.00	0.02
z_4	0.02	0.08	0.34	0.27	0.01	0.04	0.08
z_0 (e_2)	0.18	0.10	0.18	0.71	0.22	0.06	0.13

$\beta = 4$, 2 735 measured test galaxies; columns $R = R_{\mathrm{g}}$, $q = q_{\mathrm{ax}}$, $e_{1,2} = \epsilon_{1,2}$, $F = F_{\mathrm{g}}$, colour c, concentration C

Spearman rank correlation $|\rho|$, ± 1 for any monotonic relation; size R_{g}, brightness F_{g}, ellipticity ϵ_1, ϵ_2 from second moments, axis ratio q_{ax}.

- z_5, $|\rho| = 0.93$ with R_{g}, 0.97 with F_{g}
- ϵ_1 in z_7, z_9, ϵ_2 in z_0, z_2; q_{ax}, symmetric in them, $|\rho| \leq 0.13$
- colour $|\rho| \leq 0.22$; reconstructions are grey

Elongation Turns Counterfactual Spirals Edge-On



$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \delta\,\mathbf{e}_7$, decoded and classified by a CNN trained on real images.

- 249 face-on spirals; at $\delta = 3$ median q_{ax} 0.65 $\rightarrow$ 0.37, edge-on 12% (6% at $\delta = -3$)

Conditioning and Disentangling Answer Different Questions

	conditional VAE	**β-VAE**
training data	pairs $(\mathbf{x}, \mathbf{c})$	samples $\mathbf{x}$ only
loss	$\mathcal{D} + \mathcal{R}$, $\mathbf{c}$ into encoder and decoder	$\mathcal{D} + \beta\mathcal{R}$
factor	chosen and given as $\mathbf{c}$	found by the network, named afterwards
guarantee	$\mathbf{c}$ controls what it labels	none in general
use	generate on demand	find collective variables

Factors found by a β-VAE can become the labels of a cVAE.

Summary

VAE

$\mathbf{z} \sim \mathcal{N}(0, I)$, then $\mathbf{x}$ from the decoder.

ELBO, reconstruction minus a KL term.

Given $\mathbf{z}$, pixels or spins are independent.

Synthetic spins

The decoder is a mean-field ansatz.

Phase, order parameter, symmetry survive.

Energy, c, ξ, string tension are lost, less so at long range.

β-VAE

Larger β lowers the total correlation.

J_1–J_2, each order parameter gets its own coordinate at $\beta = 4$.

Trajectories, MIG 0.03–0.07 for every β; seeds matter as much as β.

Thank You

Questions, project ideas and feedback to marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar



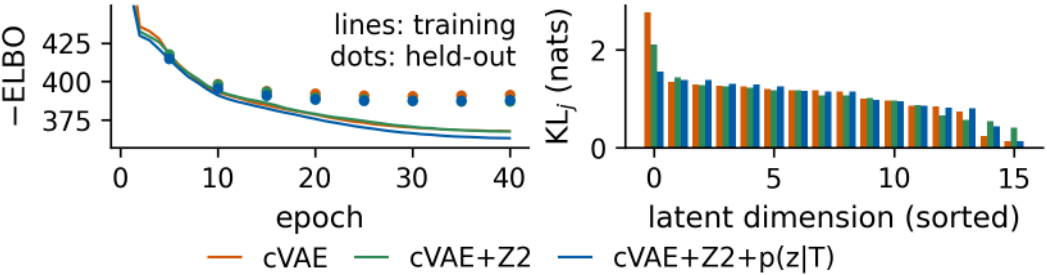
11

Backup



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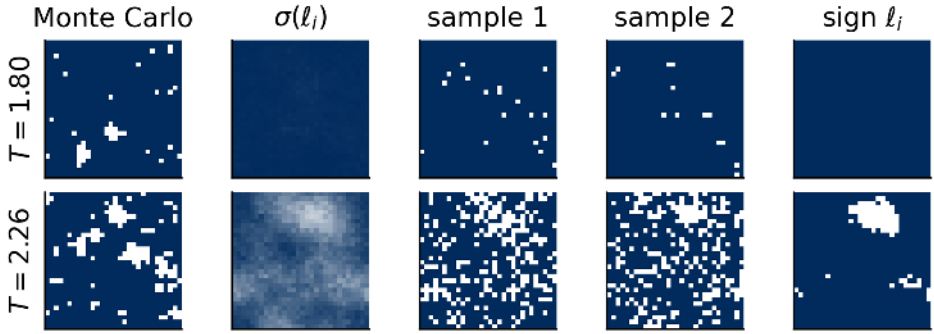
Training Health Does Not Rank the Three Spin cVAEs



Left, $-$ ELBO per configuration against epoch, training (lines) and held-out sets A, B (dots). Right, KL term of every latent coordinate, sorted.

- held-out $-$ ELBO, 391, 387, 388 nats against $N \ln 2 = 624$ for fair coins
- all 16 coordinates active; 81 % of the spins reproduced from $\mathbf{z} = \boldsymbol{\mu}$

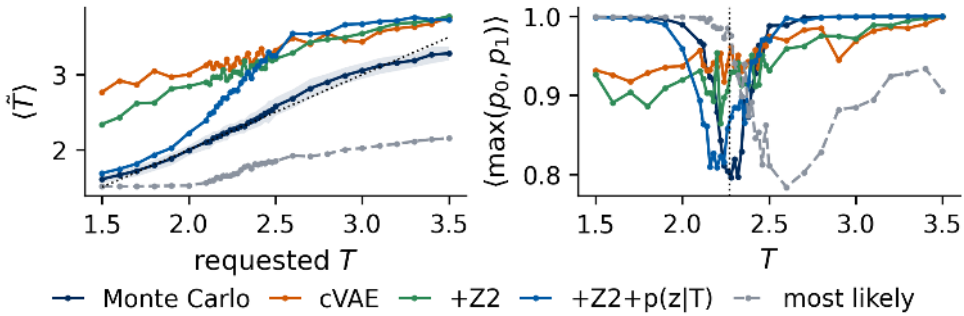
One Synthetic Configuration, Drawn from Its Probabilities



$+p(\mathbf{z}|T)$, $T = 1.80$ (top) and 2.26 (bottom). From left, Monte Carlo; sigm(ℓ_i) for one $\mathbf{z}$; two Bernoulli samples; $\operatorname{sign} \ell_i$.

- $G(1)$, samples 0.894, 0.520; independent spins 0.895, 0.521; Monte Carlo 0.930, 0.717

Networks Trained on Monte Carlo Data See the Phase



Left, temperature regressor $\langle \tilde{T} \rangle$ against T. Right, phase-classifier confidence $\langle \max(p_0, p_1) \rangle$, trained on $T \leq 1.8$ and $T \geq 2.8$.

- root-mean-square error (RMSE) of $\tilde{T}$, Monte Carlo 0.12, cVAE 0.98, $+p(\mathbf{z}|T)$ 0.62
- confidence minimum at $T = 2.28$ (Monte Carlo), 1.70 (cVAE), 2.20 ($+p(\mathbf{z}|T)$)

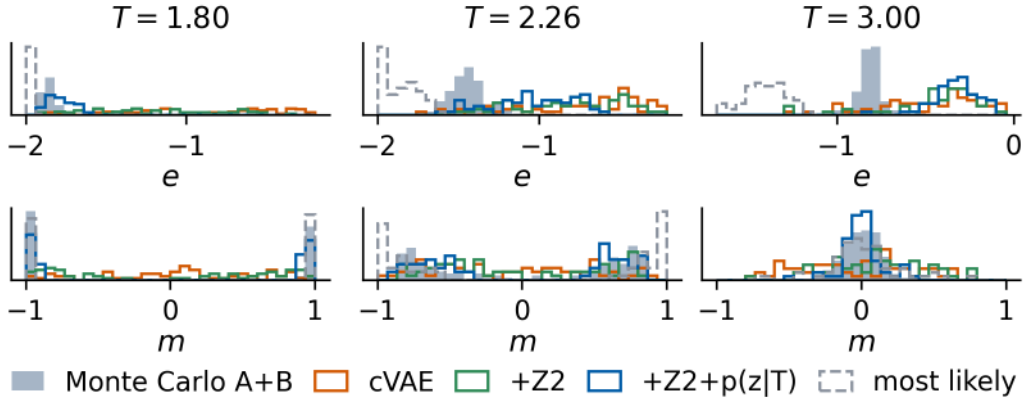
Noise Ratios of the Ising Generators

R_O over the 37 temperatures, $L = 30$; $R_O \approx 1$ means as close to set A as the independent set B.

	$\langle e \rangle$	c	$W_1(e)$	$G(L/2)$	U_4	$\langle \lvert m \rvert \rangle$	χ	$W_1(m)$
cVAE	60.0	155	38.1	15.0	2.1	13.3	11.8	3.3
+Z2	50.0	146	31.9	12.4	1.7	10.8	9.7	2.8
$+p(\mathbf{z}\lvert T)$	34.8	39.0	21.6	3.0	4.2	2.8	2.0	1.3
most likely	39.6	6.6	24.5	11.1	3.2	9.6	3.7	2.8

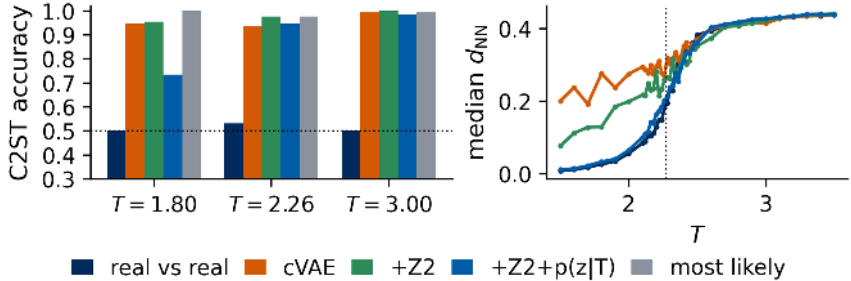
W_1, Wasserstein-1 distance of two samples, the mean absolute difference of the sorted values. Mixtures of the two ordered states below T_c, Monte Carlo 1 %, cVAE 45 %, +Z2 27 %, $+p(\mathbf{z}\lvert T)$ 1 %.

Distributions of Energy and Magnetisation



Histograms of e (top) and m (bottom) at $T = 1.80$, 2.26, 3.00; grey, Monte Carlo sets A and B. The specific heat of $+p(\mathbf{z}|T)$ is 12.54 against 1.94 at $T = 2.26$; $P(e)$ is far too broad.

A Two-Sample Test Separates Real from Synthetic

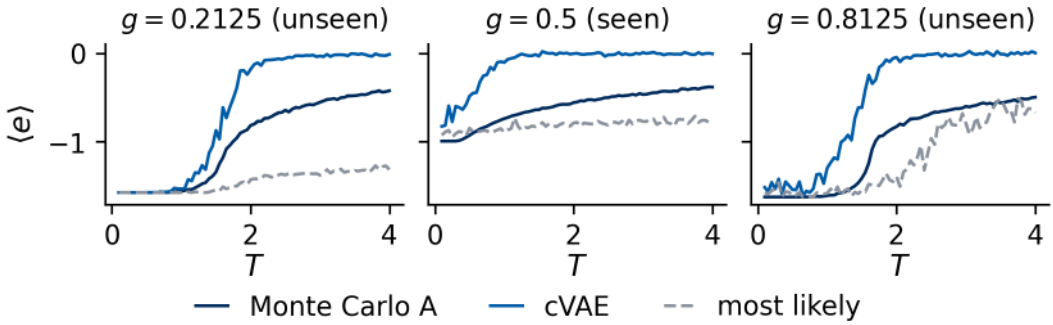


Left, classifier two-sample test (C2ST) accuracy, 0.5 means indistinguishable; right, distance d_{NN} to the nearest training configuration.

- real against real 0.50, 0.53, 0.50; $+p(\mathbf{z}|T)$ 0.73, 0.95, 0.98
- d_{NN}, Monte Carlo 0.03, 0.18, 0.43; generator 0.03, 0.20, 0.43, no memorisation

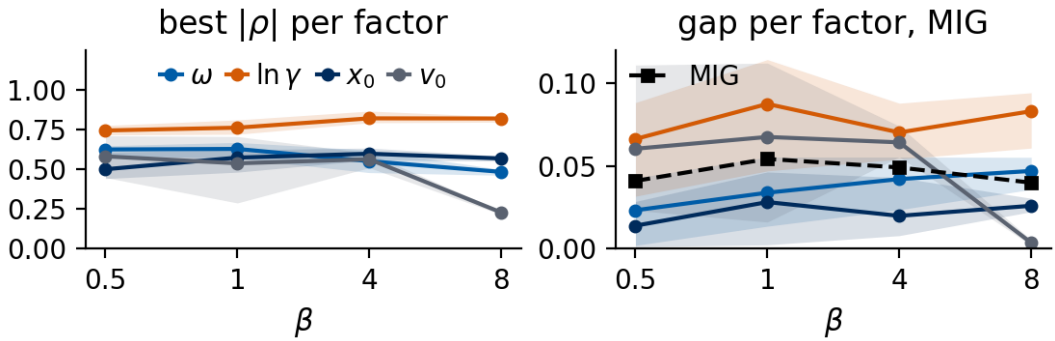
D. Lopez-Paz, M. Oquab, ICLR 2017, arXiv:1610.06545

The J_1–J_2 Energy Fails as for the Ising Model



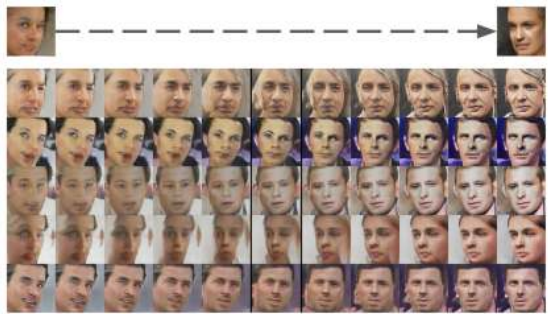
$\langle e \rangle$ against T at three values of g, two unseen and one seen. In the paramagnet the synthetic energy is close to zero, the value for independent spins; $R_e = 63.9$ at unseen points.

The MIG of the Trajectory Models Stays Small for Every β



$\mathrm{MIG} = \frac{1}{K}\sum_j [I(z_{(1)}; v_j) - I(z_{(2)}; v_j)]/H(v_j)$, the gap between the two coordinates with the largest mutual information with factor v_j, in units of its entropy. Trained models 0.03–0.07; true factors 0.92, rotated by 45° 0.46.

A Latent Vector Rotates Faces from Left to Right



Latent difference between faces looking left and faces looking right (four of each, averaged), added in steps to random generated faces (rows). The pose changes, the person largely stays. Figure 8 of A. Radford, L. Metz, S. Chintala, arXiv:1511.06434